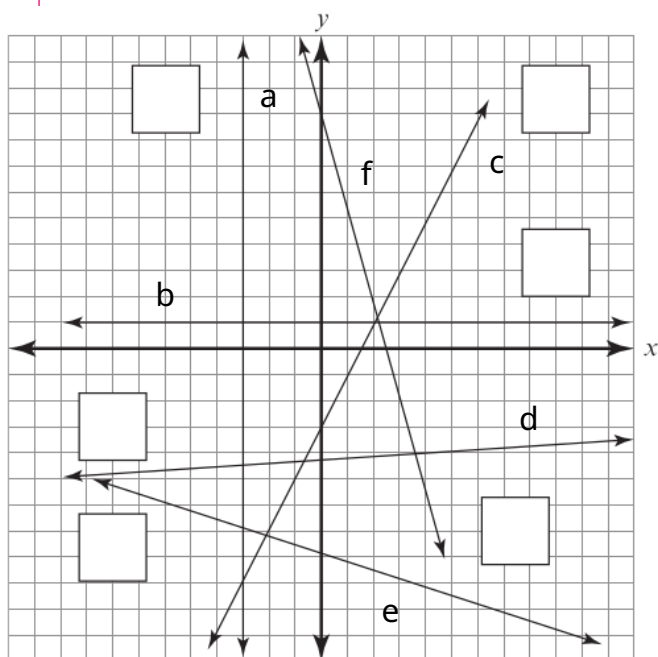


Part 1

1. Determine the slope of each of the following lines (a-f), showing ALL work.



(a) slope = undefined (b) slope = 0

(c) slope = $\frac{-1 - (-9)}{1 - (-3)} = \frac{8}{4} = 2$

(d) slope = $\frac{-4 - (-4.5)}{5 - (-3)} = \frac{1}{16}$

(e) slope = $\frac{-10.5 - (-6)}{8 - (-5.5)} = \frac{-4.5}{13.5} = -\frac{1}{3}$

(f) slope = $\frac{-2 - 9}{3 - 0} = -\frac{11}{3}$

2. On the grid below, plot the following sets of points and draw the line represented by each plot.

a)

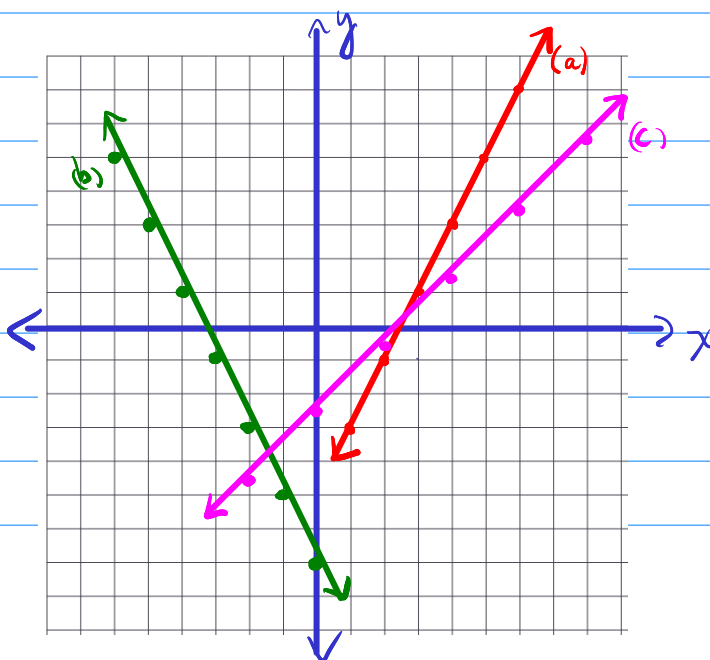
x	y
1	-3
2	-1
3	1
4	3
5	5
6	7
7	9
8	11

b)

x	y
-6	5
-5	3
-4	1
-3	-1
-2	-3
-1	-5
0	-7
1	-9

c)

x	y
-2	-4.5
0	-2.5
2	-0.5
4	1.5
6	3.5
8	5.5
10	7.5
12	9.5



$$\text{slope} = \frac{-3-5}{5-(-3)} = \frac{-8}{8} = -1$$

11. What is the equation for a straight line passing through $(-3, 5)$ and $(5, -3)$?

$$y-5 = (-1)(x-(-3)) \rightarrow y-5 = -x-3$$

$$y = -x-3+5 \rightarrow y = -x+2$$

34. Graph the equation $5x - 4y = -20$.

x-int:

$$5x = -20$$

$$x = \frac{-20}{5}$$

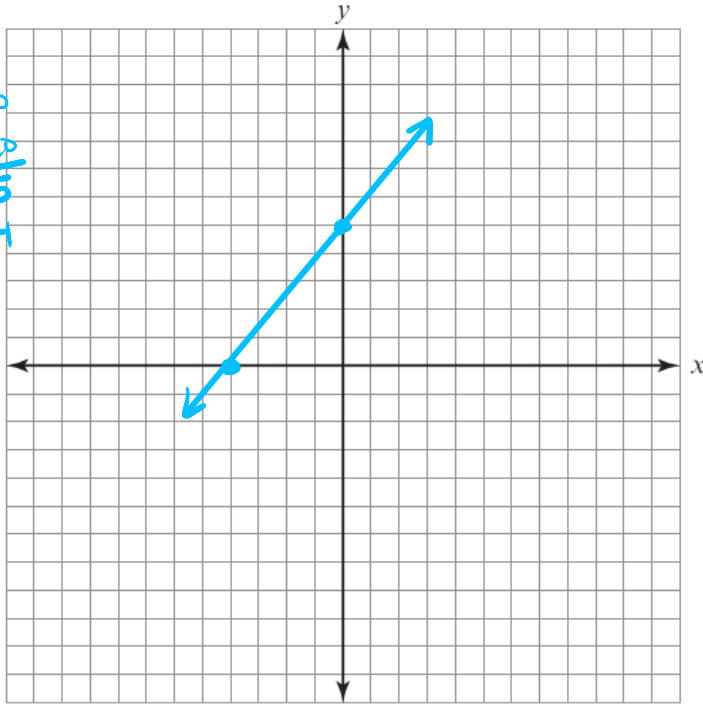
$$x = -4$$

y-int:

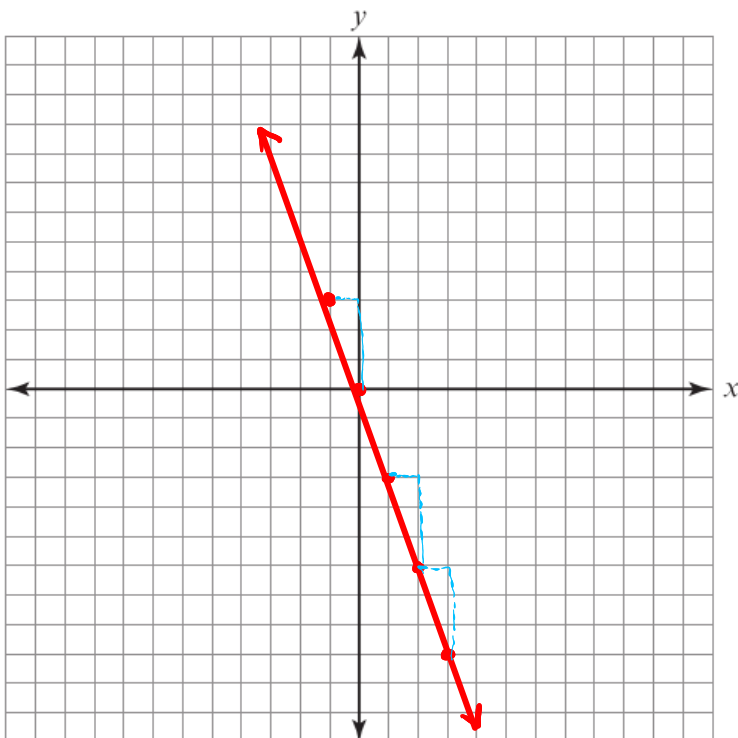
$$-4y = -20$$

$$y = \frac{-20}{-4}$$

$$y = 5$$



33. Graph the equation of a straight line that passes through $(-1, 3)$ and has a slope of -3 .



A B C D
(3,15), (-1,-14), (-2, -18) and (2,6).

7. Which of the following points is on the line $y = 7x - 4$?

$$A \ y = 7(3) - 4 = 17 \text{ (no)}$$

$$B \ y = 7(-1) - 4 = -11 \text{ (no)}$$

$$C \ y = 7(-2) - 4 = -18 \text{ (yes) } \checkmark$$

$$D \ y = 7(2) - 4 = 10 \text{ (no)}$$

48. What is the equation of a line passing through $(4, 5)$ with slope $-\frac{2}{3}$?

4. A point on the line $4x + 3y = b$ is $(3, 2)$. What is the y coordinate for $x = 4$?

$$48. \ y = mx + b$$

$$5 = (-\frac{2}{3})(4) + b$$

$$5 = -\frac{8}{3} + b$$

$$b = \frac{15+8}{3} = \frac{23}{3} \rightarrow y = -\frac{2}{3}x + \frac{23}{3}$$

$$4. \ 4(3) + 3(2) = b \rightarrow b = 18$$

$$4(4) + 3y = 18 \rightarrow 3y = 18 - 16$$

$$3y = 2$$

$$y = \frac{2}{3}$$